

1. Administrative & Study Identification

Full Study Title: A Phase 1B, Open-label Study of Elranatamab in Combination with Carfilzomib Plus Dexamethasone and Elranatamab in Combination with PF-07901801 in Participants with Relapsed Refractory Multiple Myeloma **Brief Title:** A Clinical Trial of Four Medicines (Elranatamab Plus Carfilzomib and Dexamethasone or Maplirpacept) in People With Relapsed Refractory Multiple Myeloma **Short Title/Acronym:** MagnetisMM-20 **Protocol Number:** MagnetisMM-20 **NCT Number:** NCT05675449 **Sponsor/Company Name:** Pfizer **Investigational Product:** Elranatamab, Carfilzomib, Dexamethasone, Maplirpacept (PF-07901801) **Phase of Development:** Phase 1 **Trial Status:** Recruiting **Medical Monitor:** Pfizer Clinical Trial Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of the combination of elranatamab + carfilzomib + dexamethasone (Part 1), and elranatamab + maplirpacept (Part 2), and to identify the optimal dose(s) of elranatamab + maplirpacept. **Secondary Objectives:** To evaluate treatment-emergent adverse events (TEAEs), adverse events (AEs), and laboratory abnormalities, as well as best overall response, objective response rate (ORR), complete response rate, time to response (TTR), duration of response (DOR), progression-free survival (PFS), overall survival (OS), and minimal residual disease (MRD) negativity rate.

2.2 Background & Rationale To address the unmet medical need in patients with relapsed or refractory multiple myeloma (RRMM) who have exhausted previous lines of therapy. Combining elranatamab (a bispecific antibody targeting BCMA on myeloma cells and CD3 on T cells) with carfilzomib or maplirpacept may enhance the activation of immune surveillance of multiple myeloma cells through different and complementary mechanisms, potentially driving deeper and more durable responses.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Multiple experimental cohorts, non-randomized) **Blinding:** Open-label **Randomization:** N/A (Non-randomized)

3.2 Planned Enrollment & Duration Target Enrollment (\$N\$): 90
participants Number of Sites: Multicenter **Estimated Study Start:**
14-Dec-2022 **Estimated Primary Completion:** 22-Apr-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a prior diagnosis of multiple myeloma as defined by IMWG criteria. 2. Measurable disease based on IMWG criteria as defined by at least 1 of the following: - Serum M-protein ≥ 0.5 g/dL. - Urinary M-protein excretion ≥ 200 mg/24 hours. - Serum immunoglobulin FLC ≥ 10 mg/dL AND abnormal serum immunoglobulin kappa to lambda FLC ratio (<0.26 or >1.65). 3. Part 1: Received at least 1 but not more than 3 prior lines of therapy for multiple myeloma (induction therapy followed by stem cell transplant and consolidation/maintenance therapy will be considered as 1 line of therapy). 4. Part 2: Received at least 3 prior lines of therapy and refractory to at least one IMiD, one PI, and one anti-CD38 antibody. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 6. Resolved acute effects of any prior therapy to baseline severity or CTCAE Grade ≤ 1 . 7. Not pregnant or breastfeeding and willing to use contraception.

4.2 Exclusion Criteria 1. Plasma cell leukemia, Smoldering MM, Waldenström's macroglobulinemia, Amyloidosis, POEMS Syndrome, or Primary refractory MM. 2. Active, uncontrolled bacterial, fungal, or viral infection. 3. Stem cell transplant within 12 weeks prior to enrollment, or active graft versus host disease. 4. Any other active malignancy within 3 years prior to enrollment, except for adequately treated basal cell/squamous cell skin cancer or carcinoma in situ. 5. Impaired cardiovascular function or clinically significant cardiovascular diseases. 6. Part 1: Previous treatment with a BCMA-directed therapy or prior treatment with carfilzomib. 7. Part 2: Previous treatment with any anti-BCMA directed therapy (with the exception of CAR-T) or previous treatment with a CD47-SIRP alpha-directed therapy. 8. Live attenuated vaccine within 4 weeks of the first dose of study intervention. 9. Administration with an investigational product concurrent with study intervention or within 30 days preceding the first dose. 10. Any of the following within 3 months of enrollment: erosive esophagitis, treatment-resistant peptic ulcer, infectious or inflammatory bowel disease, pulmonary embolism or uncontrolled thromboembolic event. 11. Intolerance to or severe (Grade ≥ 3) allergic or anaphylactic reaction to antibodies or therapeutic proteins.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Part 1:** Elranatamab + Carfilzomib + Dexamethasone administered over 4-week (28-day) cycles. - **Part 2:** Elranatamab + Maplirpaccept

administered over 4-week (28-day) cycles. Initial step-up dosing requires a hospitalization period for the first 2-3 doses (minimum 2 days for the first dose, 1 day for the second). **Route of Administration:** Elranatamab (Subcutaneous injection); Carfilzomib and Maplirpacept (IV infusion); Dexamethasone (Oral pill or IV). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent. Participants will take part in this study for about 2 years after the first dose.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for MM and toxicity management (e.g., step-up dosing prophylaxis). **Prohibited:** Live attenuated vaccines within 4 weeks of the first dose of study intervention. Concurrent investigational products within 30 days preceding the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline labs, Disease Assessment
Baseline / Cycle 1	Day 1 to 28	Step-up dosing (inpatient monitoring for first 2-3 doses), Safety Labs, DLT evaluation
Interim	Q28 Days	1-4 visits per 28-day cycle: Subcutaneous/IV administration, Efficacy Assessment, AE Monitoring
End of Study	Up to ~2 Years	Final Survival/Progression Follow-up, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. **Part 1:** Number of participants with dose-limiting toxicity (DLT) based on DLT evaluable participants. 2. **Part 2A:** Number of participants with DLT based on DLT evaluable participants. 3. **Part 2B:** Number of participants with DLT based on DLT evaluable participants. **Measurement Method:** Evaluated from the first dose through the end of the first cycle of combination treatment to determine safety and the recommended optimal dose.

7.2 Secondary & Exploratory Endpoints - Part 1: Number of Participants with Treatment Emergent Adverse Events (TEAE) by Seriousness and Relationship to Treatment. - Part 1: Number of Participants with Adverse Events (AE) characterized by type, frequency, and severity. - Part 1: Number of Participants with Clinically Significant Change From Baseline in Laboratory Abnormalities. - Objective response rate (ORR), complete response (CR) rate, best overall response, time to response (TTR), duration of response (DOR), progression-free survival (PFS), and overall survival (OS) per IMWG criteria. - Minimal Residual Disease (MRD) negativity rate.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient monitoring during 28-day cycles with special inpatient observation during initial step-up dosing for CRS and ICANS. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for SAEs, especially dose-limiting toxicities during Cycle 1. **Data Monitoring Committee (DMC):** Internal and independent safety review committees assessing DLTs to guide dose optimization in Part 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients receiving at least one dose) and DLT-evaluable population for primary endpoints. **Sample Size Justification:** $N=90$ participants powered appropriately for a Phase 1 dose-escalation and safety expansion methodology. **Handling of Missing Data:** Standard Phase 1 imputation techniques; safety events typically evaluated on an as-observed basis without imputation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during Cycle 1 step-up dosing, followed by regular 28-day cycle clinical visits. **Audit Trail:** Standard EDC (Electronic Data Capture) systems with strict query resolution processes for DLT and serious adverse event recording.

11. Study Identifiers & Governance

Primary ID: MagnetisMM-20 **Registry Number:** NCT05675449 **Posting ID:** 1055316693966694070 **Sponsor/Lead Organization:** Pfizer **Study Title:** MagnetisMM-20: Elranatamab Combination Therapy in RRMM **Official Regulatory Title:** A Phase 1B, Open-label Study of Elranatamab in Combination with Carfilzomib Plus Dexamethasone and Elranatamab in Combination with PF-07901801 in Participants with Relapsed Refractory Multiple Myeloma

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition: Relapsed Refractory Multiple Myeloma (RRMM) **Diagnosis Threshold:** Measurable disease per IMWG criteria **Medical Methodology:** Bispecific Antibody (BCMAxCD3) Combination Therapy **Optimization Target:** Optimal combination dosing, manageable safety profile, and maximal MRD negativity

13. Study Procedures & Methodology

Management Pathway: Relapsed/Refractory standard monitoring with mandated inpatient step-up administration
Education Protocol (HCP): ASTCT grading and management of CRS and ICANS
Education Protocol (Patient): Onsite education regarding signs of infection, CRS, and neurologic toxicity
Quality Audit Mechanism: Strict DLT period evaluation and continuous IMWG response criteria tracking

14. Observation & Follow-Up Schedule

Total Duration: Approximately 2 years (or until disease progression)
Onsite Visit Cadence: Up to 3 visits during Cycle 1; 1-4 clinic visits per subsequent 28-day cycle
Remote Monitoring Frequency: Ad-hoc based on patient symptoms and AE reporting
Checklist Review Interval: Safety and response assessments performed at the conclusion of each 28-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date		:---		:---		:---		:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]									
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]									
Statistician	[Name]	_____	[DD-MMM-YYYY]									